

Veloce and wolfSSL

Status, response, and next steps. August 27, 2026. Reference: [plan/veloce-engineering-guide.pdf v0.3](#).

1 Where each platform stands

Platform	Status
Linux x86-64	Complete. Agent, SDK, CLI, qSearch, desktop UI with entropy page, 45 gate tests green, pre-release archive <code>veloce-1.0.0-rc1-linux-x86_64</code> built. Entropy is the Lightrider seed callback (OS kernel entropy, RCT/APT verified, fail-closed). The running machine is not a listed #4718 OE; status is reported honestly and the Azure Linux OE (UG 2.2.38, OE36) awaits publication.
Windows 11 x86-64	Source complete (named pipe, restricted DLL loading, service, MSI/ZIP builder, desktop app). Not yet done: build the FIPS DLL on a Windows host per UG 2.2.45 (OE49), build the agent and PQC DLL, run the gate battery natively, sign the MSI. The FIPS claim stays pending_publication until the OE appears on the published #4718 security policy.
macOS	Desktop app and packaging (.app/DMG) work in discovery-only mode. A validated macOS agent is not possible: no wolfCrypt certificate lists any macOS operational environment (the User Guide's 51 OEs contain none). macOS ships as qSearch plus dashboard, with the dashboard honestly showing "Agent unavailable".
UI dashboards	Complete: security dashboard, qSearch page, entropy page (streaming specs plus the cloud EMS toggle). The UI performs no cryptography and reports live agent evidence only.

2 What the received documents settle

The wolfCrypt v5.2.1 FIPS User Guide (doc v1.3, 2026-03-05) and the Windows `user_settings.h` answer most of our open questions:

- Our Windows OE is already in the User Guide.** Section 2.2.45 (OE49) documents Windows 11 Pro on Lenovo ThinkPad T14 with Intel i7-1260P, built with the supplied `user_settings.h` (`HAVE_FIPS_VERSION 5 / MINOR 2 / PATCH 1`). Only certificate publication remains.
- Toolchain pin.** OE49 lists Visual Studio 2026 as compiler, linker, assembler, and debugger. The UG states a module compiled with a toolchain not on the list is not FIPS validated until retested. Our Windows build must use Visual Studio 2026.
- No PAA on OE49.** The UG marks OE49 "no PAA only" and warns PAA is not allowed without retesting. We build without `WOLFSSL_AESNI`.
- Windows integrity-hash flow.** The `fips-hash.sh` equivalent on Windows is documented: build, run once, read the hash from the FIPS callback, set it via `WOLFCRYPT_FIPS_CORE_HASH_VALUE` or `verifyCore[]`, rebuild.
- Our entropy design is the documented pattern.** UG section 6 instructs developers without RDSEED to provide their own seeding function from a strong entropy source, and the API table describes `wc_SetSeed_Cb_fips` as the hook for an external entropy source that "may be SP800-90B or may not". The Lightrider provider (OS kernel entropy plus RCT/APT verification, fail-closed) fits this exactly, under the caveat "no assurance of the minimum strength of generated keys".
- Azure Linux recipe.** UG 2.2.38 (OE36) gives the exact `configure/make/fips-hash` flow and toolchain (`gcc 13.3`) for the tested cloud OE.

3 Our response to wolfSSL

Subject: RE: FIPS #4718: User Guide received, remaining items

Thank you. The v5.2.1 User Guide and Windows `user_settings.h` answer our build questions, including the OE49 recipe, the Visual Studio 2026 toolchain, the no-PAA requirement, and the Windows integrity-hash flow. Four items remain:

1. Publication. Please notify us when OE49 (Windows 11 Pro, i7-1260P) and OE36 (Azure Ubuntu 24.04 on Docker) appear on the published #4718 certificate and security policy, and share any expected CMVP timeframe. Until then we ship Windows with validation status “pending publication”.
2. Entropy confirmation. Per UG section 6 we register our own generate-seed callback through `wc_SetSeed_Cb`: it reads OS kernel entropy (`getrandom` on Linux, `BCryptGenRandom` on Windows) and applies our own SP 800-90B-style repetition-count and adaptive-proportion checks before returning, failing closed. We document it as the CO-registered external seed source under legacy IG 9.3.A with no ESV claim. Please confirm this raises no concern.
3. Authenticode. Does signing the DLL or the installer interact in any way with the in-core integrity check we should plan for?
4. Order Form. We would like to paper: (a) Veloce as the named Program under section 3 (a PQC SDK exposing crypto services through a local agent, not a linkable crypto library); (b) commercial coverage for compiling the ML-KEM/ML-DSA sources from the public v5.9.2-stable tree as a separate non-FIPS provider, since the commercial bundle strips SHAKE; and (c) budget and timing for the FIPS v7 PQ-FS upgrade.

4 Next steps

Owner	Action	Gate
Lightrider	Provision a Windows 11 host with Visual Studio 2026; build the FIPS DLL per UG 2.2.45 (no PAA), run the integrity-hash flow, record the DLL SHA-256 into the build record.	G1
Lightrider	Build <code>veloce-agent.exe</code> and the PQC DLL; run the gate battery on Windows; package the signed MSI with the desktop app (<code>build_desktop_release.py -platform windows</code>).	G1/G4
Lightrider	Stage the Azure OE36 build (gcc 13.3, Docker on Ubuntu 24.04) as the first exactly-matching published OE for Linux deployments.	G4
Lightrider	Ship macOS as discovery-only; revisit only if a macOS OE ever appears on a wolfCrypt certificate.	n/a
Lightrider	Complete the hybrid TLS sample pair (autotools TLS+MLKEM build).	G3
Platform team	Fix <code>ems-egress/src/stream.rs</code> (corrupted at HEAD, does not compile) before S3 streaming integration.	S3
wolfSSL	Notify on #4718 SP publication of OE49 and OE36; answer items 2–4 above.	claim flip